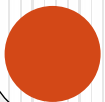
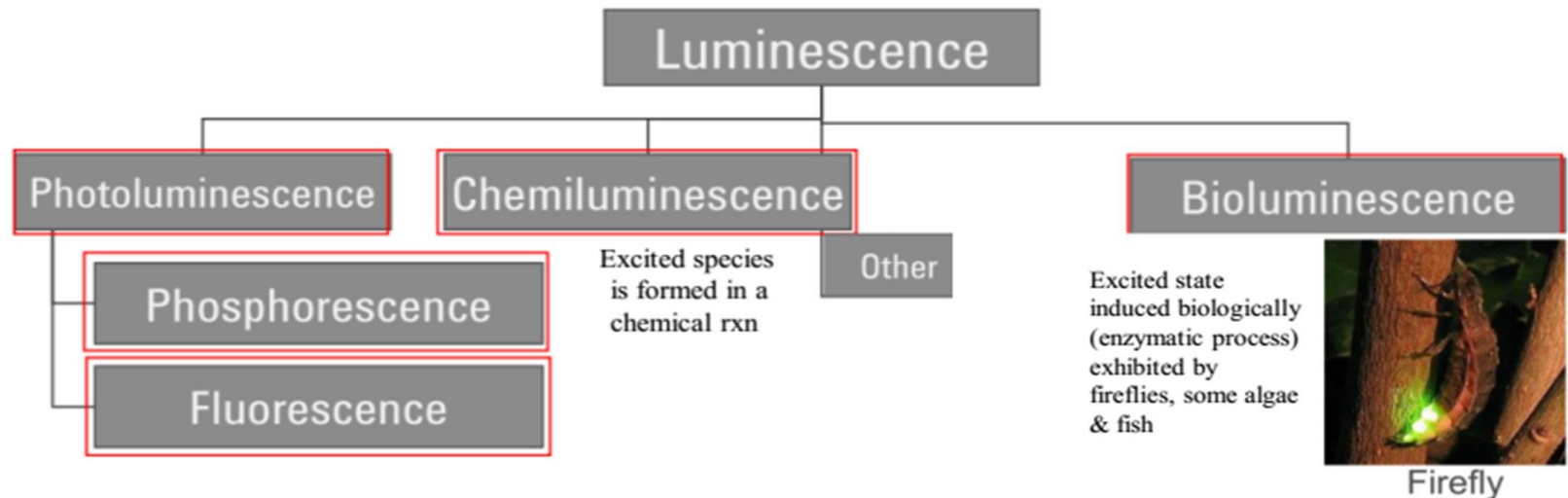


Fluorometry/Florescence Spectrophotometry



Introduction

- Luminescence is the emission of light by a substance.
- It occurs when an electron returns to the electronic ground state from an excited state and loses its excess energy as a photon.



Introduction...

- A certain chemical systems are ***Luminescence***; that is, they can be excited by EMR or chemical rxn and, as a consequence, reemit radiation either of the same or longer λ .
- In florescence & phosphorescence excitation takes place through the absorption of photons.
- In chemoluminscence an excited species is formed in a chemical rxn.
- Absorption of an ultraviolet or visible photon promotes a valence electron from its ground state to an excited state with conservation of the electron's spin.
- For example, a pair of electrons occupying the same electronic ground state have opposite spins and are said to be in a singlet spin state.

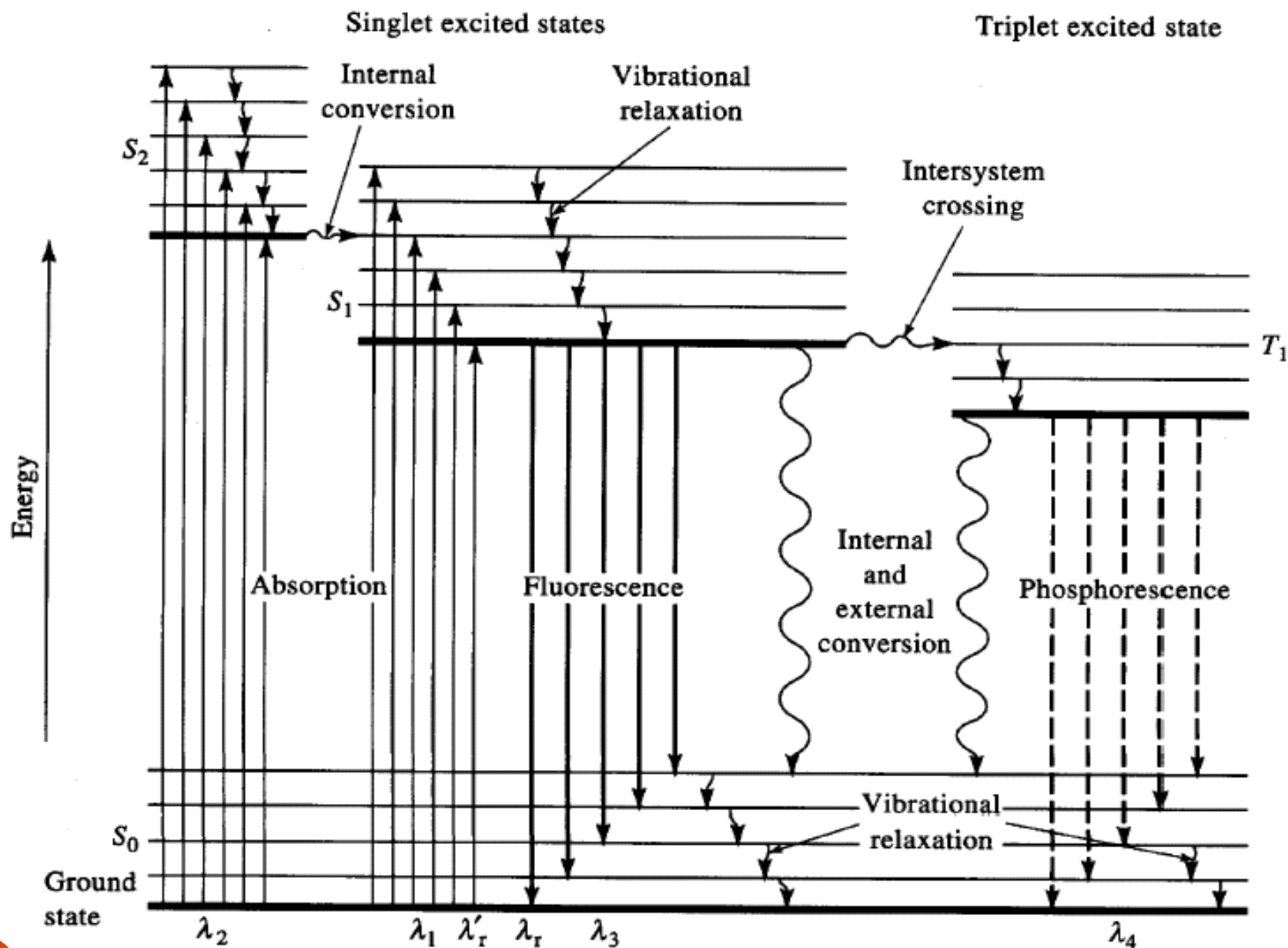


Absorbing a photon promotes one of the electrons to a **singlet excited state**.

Introduction...

- **Emission** of a photon from a singlet excited state to a singlet ground state, or between any two energy levels with the same spin, is called **fluorescence**.
- **The probability of a** fluorescent transition is very high, and the average lifetime of the electron in the excited state is only **10^{-5} – 10^{-8} s**.
- Fluorescence, therefore, decays rapidly after the excitation source is removed.
- In some cases an electron in a singlet excited state is transformed to a **triplet excited state in which its spin is no longer** paired with that of the ground state.
- Emission between a triplet excited state and a singlet ground state, or between any two energy levels that differ in their respective spin states, is called **phosphorescence**.
- **Because the average lifetime for phosphorescence** ranges from **10^{-4} – 10^4 s**, phosphorescence may continue for some time after removing the excitation source.





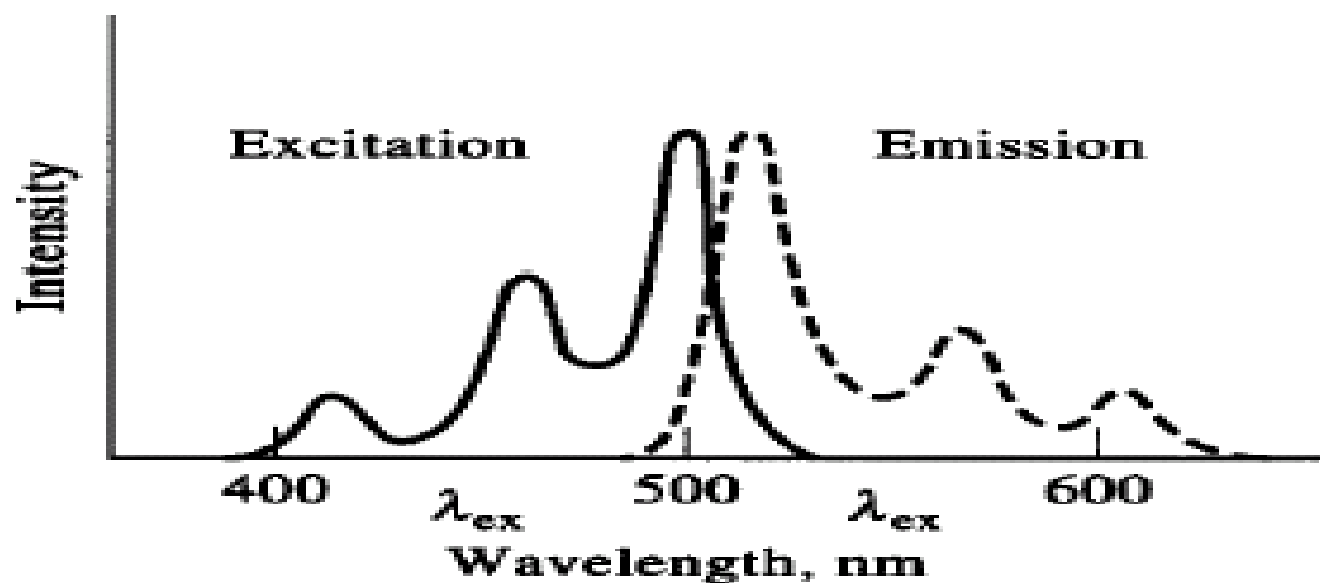
FLUOROMETRY

- Fluorometry is an analytical method , w/c utilizes the ability of some sub to exhibit fluorescence.
- Light of a λ d/t from the irradiating light is released or emitted ff the absorption process.
- Most often the irradiating light is UV light and the emitted light is visible light.
- The fluorescence spectrum of a molecule is, ideally, a mirror image of the longest wavelength band in the absorption spectrum of the molecule but often the spectrum is distorted due to partial overlap b/n the absorption & the emission spectra.
- In a molecule containing a number of UV absorption bands, the longest λ_{max} is the one associated most strongly with the production of fluorescence.



FLUOROM.....

- The shape of the fluorescence spectrum is independent of the wavelength used for excitation

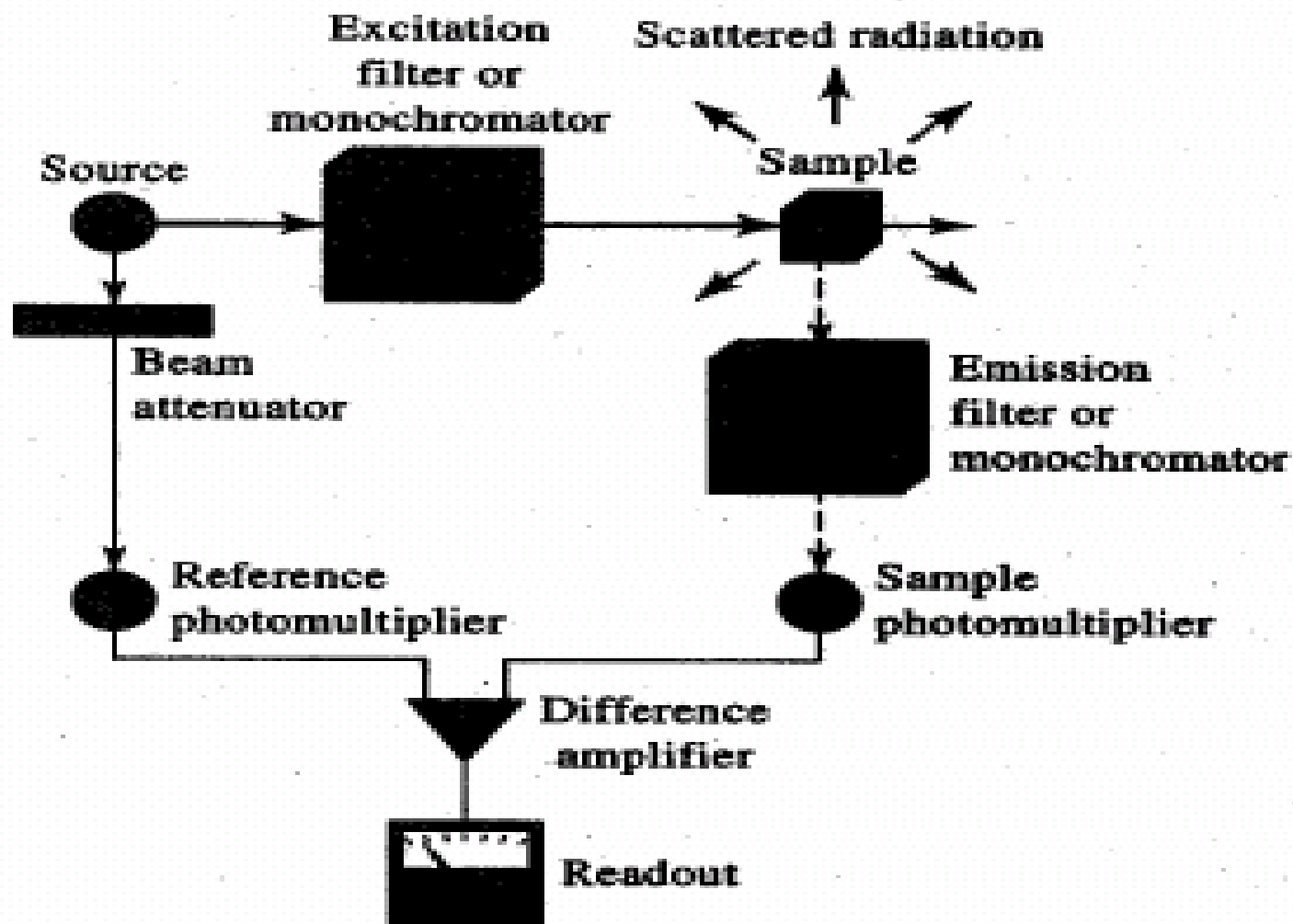


Instrumentation

- Instruments for measuring fluorescence contain the same basic components as UV/Vis spectrophotometers.
- The only d/c is presence of additional wavelength selector or dispersion device after sample cell.
- The dispersion can be achieved with filters or with Monochromators.
- Instruments using filters are generally called **fluorometers**, while instruments equipped with Monochromators are called **spectrofluorometers** since they provides both excitation and emission spectra.
- It is most convenient to measure the fluorescence at a right angle to the excitation beam to decrease the scatter from the cell walls and so/n & avoid any interference from the transmitted light from the light source.



FLUOROM....



Instrum...

Source

- The signal produced by an analyte is proportional to the number of excited analyte molecules formed per unit time.
- An intense continuum source used in most commercial fluorometers is the xenon arc lamp.

Wavelength selectors

- Two monochromators; one to select the wavelength to be used for excitation of the sample, the other to scan the wavelength range of the light emitted by the sample.

Detector

- A key requirement for a detector is its ability to detect weak optical signals.
- A photomultiplier tube is used as the detector in most fluorescence spectrophotometer.



- The main analytical application of molecular spectrofluorometry is detection & quantification of species present at concentration so low that most other techniques are not useful.

$$F = 2.3\Phi\epsilon bCP_0$$

- Φ = quantum efficiency = # molecules emitting / total # molecules excited
- ϵ (L/mol-cm) molecular absorptivity coefficient and b (cm) path length
- P_0 is incident radiant power density (watts/cm²).
- These three determine the sensitivity of fluorometry for the analyte.
- The measurement situation in fluorometry distinguishing b/n a small signal and zero signal is more favorable than that encountered in absorption spectroscopy (measuring a small d/c b/n two large numbers).



Advantages

A. Sensitivity

- Substances that are reasonably fluorescence may be determined at concentration up to 5000 times lower than those required for absorption of spectrophotometer
 - In Spectrofluorimetric measurement the detector measures single light intensity w/c may be amplified electronically many times without introducing significant noise

B. Selectivity

- Not all substances that absorb in the uv-vis fluorescence
- Wavelength of excitation /emission can be easily varied to selectively measure the fluorescence

Factors that affecting the degree of fluorescence

Structural rigidity

- It is found empirically that fluorescence is particularly favored in molecules that possess rigid structures.
- One part of a non-rigid molecule can undergo low-frequency vibrations with respect to its other parts; such motion undoubtedly account for some energy loss.



Factors....

Solvent effect

- Variation in viscosity will cause variation in the frequency of collision b/n molecules
 - Decrease in viscosity ===== decrease fluorescence by deactivation of the excited molecular collision
- If the excited state of a polar molecule has a higher dipole moment than its ground state (most molecules are in this class), the excited state will be more stabilized by interaction with a polar solvent than will the ground state.



Factors....

- As a result, upon going from a less polar to a more polar solvent, the fluorescence spectrum will shift to longer wavelengths.
- In a few cases, the ground state of a solute is more polar than the excited state.
- In this case, going to a more polar solvent stabilizes the ground state more than the excited state, causing a shift to shorter wavelengths with increasing solvent polarity.



Factors....

Temperature effect

- Variation in temperature will cause variation in the frequency of collision b/n molecules
 - Increase T° ===== decrease fluorescence by deactivation of the excited molecular collision
- Higher T° === increase thermal motion ===== deactivation through heat
- A rise in 1°C results in decrease in intensity of fluorescence by 1%



Factors....

Quenchers

- *Quenching* is the process whereby emission from excited molecule is decreased by energy transfer to another molecule (quencher)
- The emission of light by photoexcited luminescent molecules may be decreased or even eliminated by interactions with other chemical species.
- E.g. presence of dissolved oxygen



Factors....

- Two kinds of quenching:

Static quenching - complexation b/n the potentially luminescent molecule & the quencher takes place in the ground state.

- The complex, when excited, fails to luminesce.
- The efficiency of quenching is governed by the formation constant of the complex as well as by the conc of the quencher.
 - Eg. the quenching of the fluorescence of doxorubicin by Fe(III).

Dynamic quenching - the quenching species and the potentially luminescent molecule react subsequent to photoexcitation of the latter and during the lifetime of its excited state.

- Its efficiency depends on the viscosity of the solution, the lifetime of the excited state of the luminescent species, and the conc of the quencher.



Factors....

- The fluorescence of a molecule is decreased by solvents containing heavy atoms or other solutes with such atoms in their structure e.g. carbon tetrabromide and ethyl iodide.
- Heavy atoms in solution quench fluorescence by colliding with excited molecules so that their energy is dissipated, i.e. chloride or bromide ions in so/n cause collision quenching

Oxygen

The presence of oxygen may interfere in two ways

- By direct oxidation of fluorescence substances to non-fluorescence products
- By Quenching of fluorescence
- De aerated solution



Factors....

Concentration Effect

- If the conc of a so/n prepared for fluorescence measurement is too high, some of the light emitted by the sample as fluorescence will be reabsorbed by other unexcited molecules in so/n.
- For this reasons, fluorescence measurement are best so/n as a sample 10-100 weaker than those w/c would be used for measurement by UV spectrophotometry

Effect of pH

- Fluorophores that contain ionizable groups are affected by the pH
 - The fluorescence of an aromatic cpd with acidic or basic ring substituents is usually pH dependent.
 - Fluorescence intensity from excited states of charged and uncharged species is generally different.
(i.e. change in pH alter the ratio of charged and uncharged species)
 - Both the λ and the emission intensity are likely to be d/t for the ionized forms of the cpd.
- The presence of dissolved oxygen often reduced the intensity of fluorescence in a so/n.

Factors...

- It is not entirely possible to predict how strongly fluorescent a molecule will be.
- For example adrenaline and noradrenaline differ in their structure by only a single methyl group but noradrenaline exhibit fluorescence nearly 20x intensely than adrenaline
- Generally, fluorescence is associated with an extended Chromophores/Auxochrome system and rigid structure
- Quinine is an example of a strongly fluorescent molecule as might be expected from its extended chromospheres and rigid structure
- The Chromophores in Ethinylestradiol is just an aromatic ring but the presence of a phenolic hydroxyl group in combination with rigid ring structure in the rest of the molecules render it fluorescent



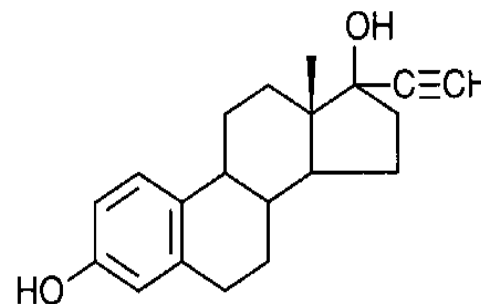
Application of fluorescence spectrophotometry in pharmaceutical analysis

- Determination of fluorescent drugs in low dosage formulation
- Limit of impurity (Fluorescent or converted fluorescent cpd)
- Determination of small amounts of drugs in biological fluids



Application of fluorescence spectrophotometry in pharmaceutical analysis...

- **Determination of dissolution rate of digoxin tablet**
- **Determination of stability of peptide drugs in so/n**
- **Determination of aluminum in water for injection as a fluorescent complex**
- **Determination of Ethinylestradiol tablets**
 - *The BP utilizes a fluorescence assay to determine Ethinylestradiol in tablets*
 - *The tablet contain low dosage of the drug so that interference by excipients is likely to cause a problem in UV/Visible spectrophotometric measurement*
 - *The sample is measured using an excitation λ of 280nm and measuring the emission at 320nm*



Ethinylestradiol



Application...

- *After the fluorescence of the sample extract in methanol has been determined, 1M NaOH so/n is added to the sample so/n and the fluorescence is determine again*
- *The addition of NaOH removes the fluorescence by ionizing the phenol groups of the Ethinylestradiol and thus any residual fluorescence w/c is due to excipients can be subtracted from the reading*
- *In the BP assay the Ethinylestradiol content of the tablet extract is determined by comparison with the fluorescence of a so/n containing a known amount of Ethinylestradiol standard analysed using the same condition*

